

SHRUTI JAIN

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EDUCATION

Carnegie Mellon University | Master of Science in Computer Vision Aug 2025 - (Expected) Dec 2026
Coursework: Visual Learning and Recognition, Advanced Computer Vision, Introduction to Machine Learning
CGPA : 4.11/4

Manipal Institute of Technology | Bachelor of Technology in Information Technology 2019-2023
CGPA : 8.71/10

SKILLS

Programming Languages/Tools: Python, C++, Java, Docker, Kubernetes, SQL
ML and CV Frameworks/Libraries: PyTorch, OpenCV, ROS, CUDA, scikit-learn, TensorFlow, Matplotlib
AI Topics: Computer Vision, Transformers, VLMs, Building Agents, Generative Models, Object Detection, Image Segmentation, Zero-shot Learning, Semi-supervised Learning, Domain Adaptation

RESEARCH EXPERIENCE

AIRLab - Carnegie Mellon University | *Research Assistant* Jan 2026 - April 2026

- [\[Project Website\]](#) Developing a multimodal medical triaging pipeline for the [DARPA Triage Challenge](#) using RGB and thermal UAV imagery to predict 50+ hierarchical medical labels, under the guidance of **Prof. Sebastian Scherer**.
- Built a two-stage system combining conditional diffusion-based RGB to IR generation and VLM fine-tuning with LoRA/GRPO optimization to improve triage reasoning under limited paired RGB-IR data.

Carnegie Mellon University, USA | *Research Assistant* Aug 2025 – Dec 2025

- Worked under **Prof. Deva Ramanan** on Developing DetPO, a gradient-free black-box prompt optimization framework achieving up to 9.7% improvement over prior approaches on Roboflow20-VL and LVIS.

Robotics Research Centre (RRC) - IIIT Hyderabad | *Research Assistant* June 2023 – Sept 2023

- Achieved **14.4%** average decrease in the mean absolute scores across 3 datasets as compared to the SOTA for **Image Segmentation in self driving cars** in adverse weather conditions.
- Implemented a **differentiable visual and latent prompting mechanism** to finetune on **Segment Anything Model** via **adaptors** through learnable image pre-processing blocks.

Nest Labs - Worcester Polytechnic Institute, USA | *Research Intern* June 2022 - Jan 2023

- Laid the ground work for **increasing the efficiency of Ant Colony Optimization** for solving the Travelling Salesman Problem by learning better features with Deep Learning using Python.

RELEVANT PROJECTS

- **Gated Multimodal Web Navigation Agents** - [\[Poster\]](#) A gated multimodal web navigation framework combining MindAct and Qwen2-VL on the Multimodal Mind2Web benchmark, improving robustness to adversarial popup attacks.
- **Editing 3D Gaussian Splats** - [\[Slides\]](#) Edits **3D Gaussian Splats** based on text prompts by using **Diffusion** based 2D Image Editing techniques.
- **Flash Attention for ViT** - [\[Poster\]](#) Implemented FlashAttention-1 and 2 in Minitorch as custom CUDA autograd ops with tiled SRAM compute and online softmax.
- **Real-time Failed 3D Print Detector** - Detects failed 3D prints in real time using **OpenCV** and anomaly detection.
- **Car Localization** - Worked on **sensor fusion and localization of a self-driving car**, given a 3D point cloud of its environment from a LiDAR.

SELECTED PUBLICATIONS

- DetPO: In-Context Learning with Multi-Modal LLMs for Few-Shot Object Detection [\[Paper\]](#) [\[Website\]](#) [\[Code\]](#) **Under Review (Rebuttal) — ECCV**
- DiffPrompt: Differentiable Implicit Visual Prompts for Semantic-Segmentation in Adverse Conditions [\[Paper\]](#) [\[Project Page\]](#) [\[Code\]](#) **Published in IROS 2024**
- Light Weight Character and Shape Recognition for Autonomous Drones [\[Paper\]](#) **Workshop acceptance in ICML 2022**
- A Studios Approach To Semi-Supervised Learning [\[Paper\]](#) [\[Code\]](#) **Workshop acceptance in NeurIPS 2021**

INDUSTRY EXPERIENCE

Palo Alto Networks | *Software Developer* Oct 2023 - April 2025

- Worked on features and APIs of the microservices (**JAVA Spring Boot**) for ingesting events from tenant's cloud accounts (**AWS, Azure, GCP**) for near real-time threat detection using **Artificial Intelligence**.
- Integrated a **scheduler-worker service** that schedules jobs to poll cloud account for events with a **job management framework** based on **SQS Queue** and **Redis cache**.